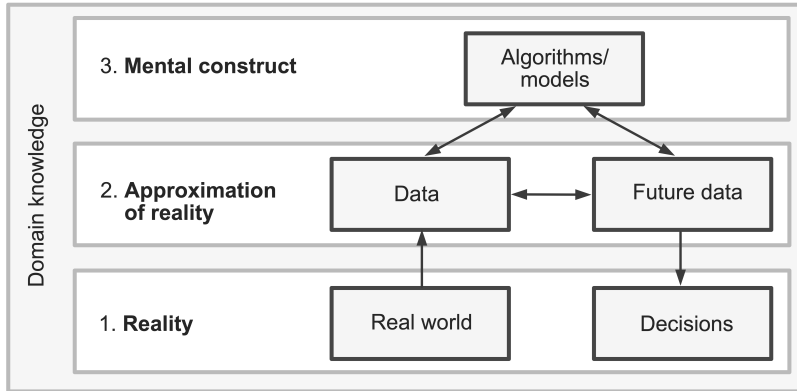


Model diagnostics

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



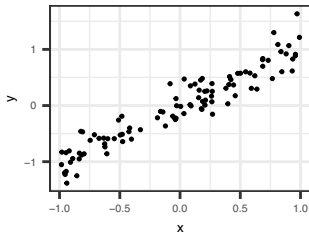
Question: Is our model useful?

Limitations of empirical logit plots

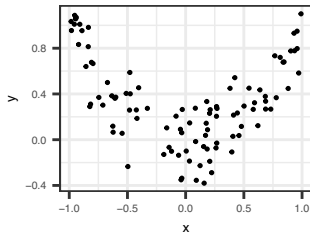
- ▶ Does not assess a fitted model (performed *before* fitting the model)
- ▶ One point per *bin*, not one point per *observation*
- ▶ Can only plot one or two explanatory variables at a time

Why residuals in linear regression are nice

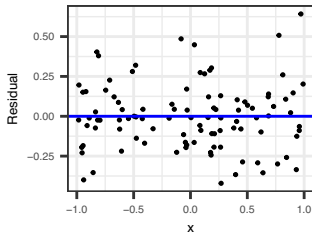
Dataset 1



Dataset 2

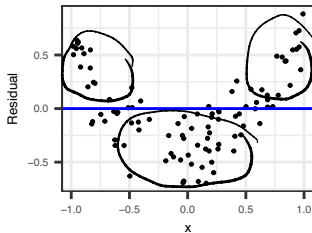


Residuals 1



↑
random scatter around 0

Residuals 2



← clear
pattern

Randomized quantile residuals

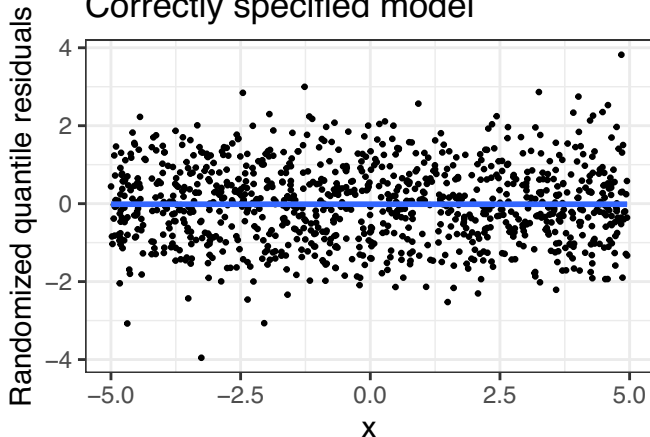
Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

quantile residuals:
a different type
of residual (not $y - \hat{y}$)

that behaves like
residuals in a
linear model

Correctly specified model

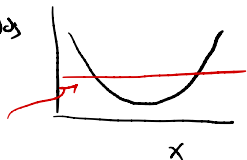


Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

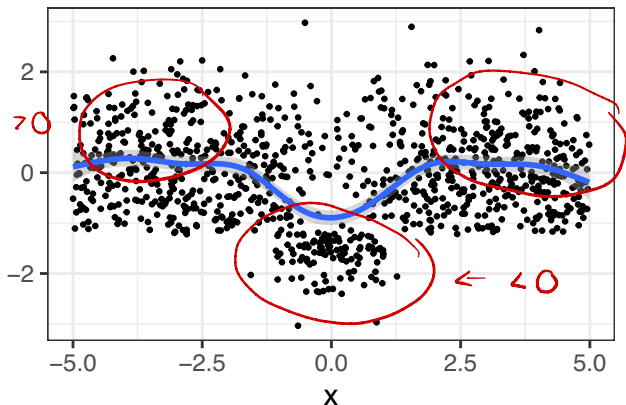
true
log odds



fitted
log odds

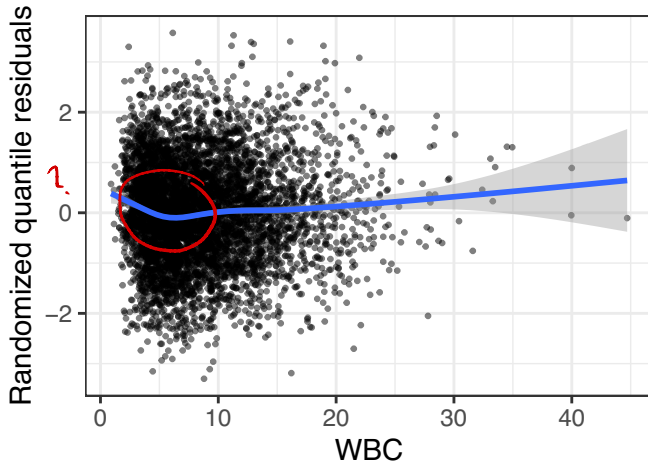
Mis-specified model

Randomized quantile residuals



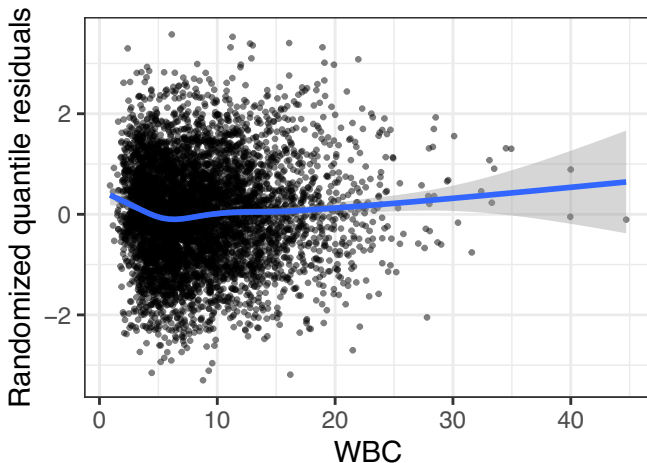
Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Question: How can we decide whether a plot is consistent with a correctly-specified model?

Parametric bootstrap

The *parametric bootstrap* generates data from the *fitted* model:

• Given original data $(X_1, Y_1), \dots, (X_n, Y_n)$, fit model
 $\rightarrow \hat{\pi}_1, \hat{\pi}_2, \dots, \hat{\pi}_n$

• simulate new responses $Y_1^*, \dots, Y_n^*$
 $Y_i^* \sim \text{Bernoulli}(\hat{\pi}_i)$
 $\uparrow$ estimated from original model!

(now we know the model is correct for Y^*)

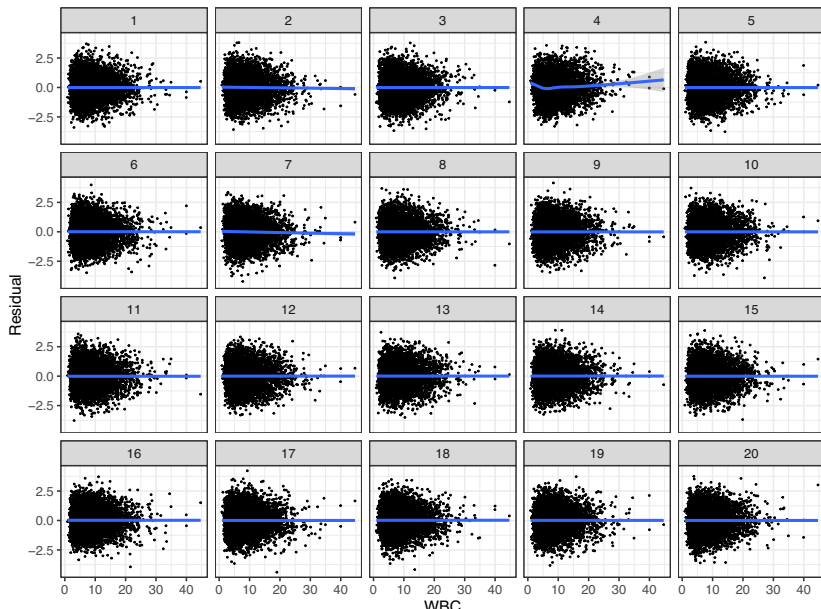
• refit model / diagnostics / etc on
simulated data $(X_1, Y_1^*), \dots, (X_n, Y_n^*)$

Class activity

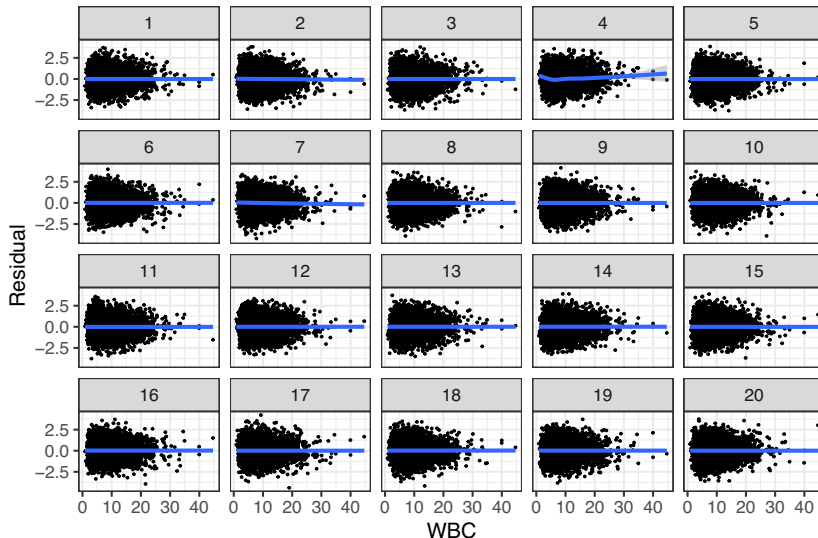
- ▶ Work on the class activity in R (linked on Canvas and the course website)
- ▶ You are welcome to work in groups
- ▶ At the end of class, submit HTML file on Canvas (one per group, listing all group member names)

Visual inference

```
## decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```



Visual inference



```
decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```

```
## [1] "True data in position 4"
```